

A Comparative Analysis of Vision-Based and GPS-Based Speed Estimation Algorithms for Autonomous Vehicles

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Abstract—Accurate estimation of vehicle speed is fundamental to autonomous navigation. This study critically evaluates the precision of a vision-based Dense Optical Flow (Farneback) algorithm, augmented with dynamic affine calibration, in comparison to consumer-grade GPS trajectory data. Utilizing the comma2k19 dataset, both methodologies were assessed against a 100Hz CAN-bus ground truth within a dynamic urban driving environment. A quantitative approach was employed, involving the calculation of Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Bias Error (MBE). The findings definitively refuted the initial hypothesis: the GPS-based approach considerably outperformed the vision system, recording an MAE of 1.68 km/h in contrast to 4.50 km/h. Whilst GPS demonstrated considerable robustness across varying environmental conditions, the pure pixel-tracking vision method proved exceptionally vulnerable to visual noise, such as shadows, and failed to reliably detect deceleration. The study concludes that standard GPS probe data is markedly superior to Dense Optical Flow for vehicle speed estimation. Future research should focus on Sensor Fusion techniques, integrating deep learning models (such as YOLO) and Kalman filtering, to address the inherent limitations of both technologies.

I. INTRODUCTION

A. Theme & Topic Rationale

This research focuses on the theme of Intelligent Transport Systems (ITS) and Autonomous Vehicle Sensor Perception. Getting an accurate estimate of a vehicle's speed is super important for safety, especially for Advanced Driver Assistance Systems (ADAS) and autonomous navigation. While GPS is widely used, they frequently encounter signal degradation in urban canyons, tunnels, or adverse weather conditions, which can cause errors due to signal reflections (multipath errors) [1], [2]. On the other hand, estimating speed using dashcam footage is a cheaper, sensor-rich option [3], but it is inherently vulnerable to variations in lighting and computational latency. It's important to compare these two methods against a highly accurate baseline, specifically vehicle CAN-bus data. Such comparison elucidates the specific environmental limitations of each system and offers empirical evidence to inform the development of improved algorithms in the future.

B. Positioning & Research Onion

Based on Saunders' Research Onion [4], this research uses positivism, meaning seeing reality as something objective and separate from what other people think or interpret. The data that will be used is mainly measurable data like speeds, GPS coordinates and pixel shifts, with the researcher acting as an impartial observer by analysing the physical data. This research takes a deductive approach starting with already proved formulae and concepts such as the Haversine formula for GPS speeds and ego-motion tracking and comparing the data with each other to see which is more accurate. Even though the data comes from an archived source (Comma2k19) [5], the main objective for this research is to compare two algorithms under controlled conditions, using a steady CAN-bus ground truth to see how much error each algorithm has. Methodologically, the data is all quantitative because we are turning visual and geospatial information into numbers to analyse and compare statistically meaning there isn't any qualitative data involved. The analysis is cross-sectional, so we are looking at a specific snapshot of data to test how well the algorithms work now, rather than tracking changes over time. Finally, we are collecting secondary data like video clips and GNSS logs to analyse and find the Mean Absolute Error for both algorithms.

C. Background To This Research Theme

The push towards fully autonomous vehicles has sped up the need for robust environmental awareness. In the past, navigation and speed checks have mostly relied on standard GPS receivers. But since self-driving technology needs accuracy to be down to milliseconds, the usual GPS signals and their delays or noise are not good enough [1]. Currently, recent advances in computer vision have made it possible to figure out "ego-motion", basically how a camera moves through 3D space by tracking features [3]. GPS can fail under bridges and cameras don't work well in the dark or when there is lots of glare due to these drawbacks relying on just one sensor isn't ideal thus the industry is leaning more towards combining different sensors. To manage to incorporate that

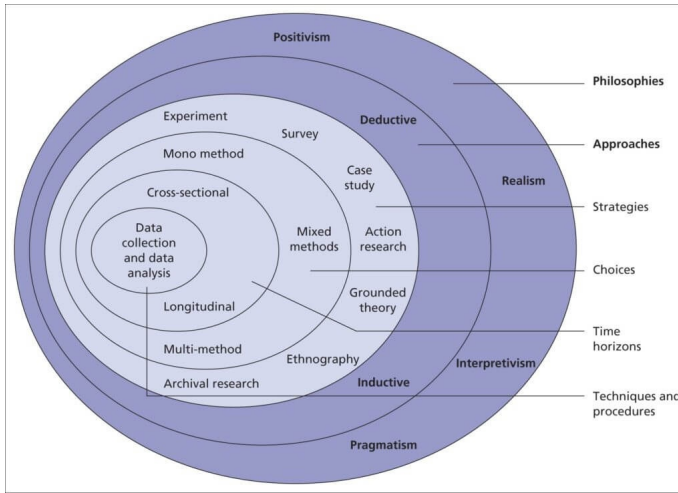


Fig. 1. Research Onion [4]

well, researchers need to first understand how accurate each sensor is on its own and what kind of errors they might have when tested on real-world situations.

D. Hypothesis

This research works with a clear null hypothesis, which basically says there is no real difference in how accurate the vision-based algorithm is compared to GPS when it comes to estimating the vehicle's true ground speed. On the other hand, the alternative hypothesis suggests that the vision-based method will have a smaller error margin than GPS during normal driving, mainly because commercial GPS units tend to have some delay and noise in their readings.

E. Research Aim & Purpose Statement

The main goal of this study is to compare how accurately a vision-based speed estimation algorithm works versus a GPS-based one, using real-world dashcam footage. Ultimately, we are trying to spot the differences in how both methods measure speed. By testing against a solid ground truth, we hope to get useful insights into how reliable these sensors are, which could help in building safer autonomous vehicle navigation systems.

II. REVIEW OF RESEARCH METHODOLOGY

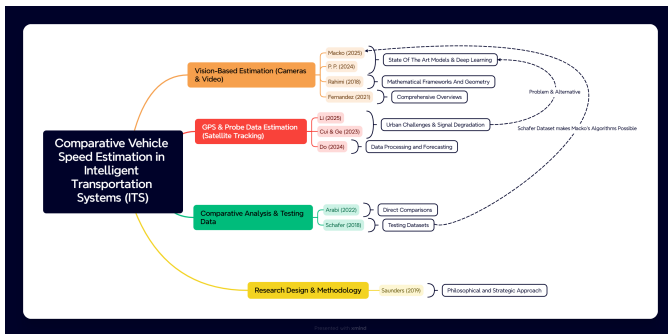


Fig. 2. Literature Map

In the field of Intelligent Transportation Systems (ITS), the selected methodology fundamentally determines the accuracy of speed estimation. Looking into this research area, you can spot two main methods. Researchers focusing on vision-based speed estimation usually take an experimental, deductive approach based on positivism. Their methods generally entails the collection of primary data through edge-computing cameras, utilising geometric transformations (such as Inverse Perspective Mapping) or pixel-tracking algorithms to ascertain speed based on observable physical displacement.

Conversely, researchers looking at GPS probe data often go for a second, more numbers-driven way of analysing things. Their methodology utilises extensive datasets comprising historical, crowdsourced vehicle trajectories. Instead of conducting real-time physical experiments, their focus is predominantly on statistical modelling, noise mitigation, and using map-matching algorithms to fix those multi-path errors that pop up in city areas.

A. Testing Strategies in the Chosen Context

To implement these methods, a review of the literature identifies three principal testing strategies utilised by researchers to evaluate speed estimation frameworks.

The initial approach is what they call Controlled Closed-Circuit Testing, whereby researchers typically initiate testing within controlled, sterilised environments. They drive their specially equipped vehicles at set speeds around a closed track. This method effectively removes extraneous variables such as traffic occlusion or unpredictable civilian driving behaviour, thereby facilitating a concentrated focus on the mathematical principles underpinning the vision or GPS algorithms.

The second approach is known as 'In-The-Wild' Highway Deployment. To check how well it scales, researchers put cameras on busy motorway overpasses. This testing method relies on obtaining a substantial sample size of uninstrumented civilian vehicles. The speed info from the vision system is then compared with big-picture GPS data from third-party companies to see how accurate it is in real-world conditions and how much delay there might be.

The third method involves Synthetic Simulation (Digital Twins). Researchers are progressively utilising advanced rendering engines to artificially produce precise optical flow fields and pixel displacement data in fake extreme weather scenarios that would be hard to mimic in real life. This methodology serves as an effective means of advancing the mathematical boundaries of speed estimation algorithms.

B. Experimental Variables and Hypothesis Validation

No matter what testing approach you use, researchers need to carefully control and tweak certain experimental variables to prove their ideas. When it comes to independent variables in vision-based studies, they often mess around with the camera angle (pitch and yaw), how far away the camera is from the target, and things like lighting and weather to see how solid the system is. Conversely, in studies involving GPS technology, the primary independent variable manipulated is the 'polling

rate’, defined as the frequency at which the GPS receiver records a coordinate.

Accompanying these inputs, the universally measured dependent variable across all studies is the calculated velocity, expressed in kilometres per hour (km/h) or metres per second (m/s), along with the error margin, which they usually measure using Root Mean Square Error (RMSE) or Mean Absolute Percentage Error (MAPE).

Furthermore, to institute control variables and prove an alternative hypothesis, such as the proposal that a vision-based model shows greater accuracy than conventional GPS. This is usually done by arming test vehicles with super-calibrated Real-Time Kinematic (RTK) GPS or using LiDAR speed guns. Finally, by isolating these variables, researchers can definitively discover the specific parameters under which their speed estimation models remain valid.

C. Distinguishing Between Academic and Non-Academic Material

When we’re looking at these variables and methods, it’s important to clearly distinguish between well-tested academic research driven by the industry that isn’t academically reviewed. Academic material sourced from peer-reviewed journals, these studies offer transparent methods, publicly accessible datasets (such as BrnoCompSpeed), and standard error metrics. They candidly acknowledge limitations, including camera calibration drift and GPS multipath errors.

Non-academic material, such as industry white papers and promotional reports from ITS hardware manufacturers, offer valuable insights into real-world adoption. However, they often lack methodological transparency. Frequently, they present “best-case scenario” accuracy figures without sufficiently detailing the experimental constraints, thereby rendering them unsuitable for establishing a scientific baseline.

D. Recommended Peer-Reviewed Articles

The following five peer-reviewed articles establish the methodological foundation for the comparison of vision and GPS techniques, each representing distinct layers of Saunders’ Research Onion.

[6] operate firmly within a positivist paradigm, employing an experimental strategy through the employment of YOLO object detection in conjunction with vanishing point geometry. This study exemplifies contemporary standards for quantitative methodologies and procedures, avoiding reliance solely on fixed road-width assumptions.

Additionally [7] offer a highly pertinent comparative methodology in their work “Comparison of the GPS and Camera Distance Estimation”. Their research demonstrates a deductive approach, establishing a rigorous framework for experimental validation by utilising RTK GPS as an infallible ground truth to test alternative hypotheses against monocular camera estimations.

Furthermore, [8] is integral to its archival research strategy, delineating the methodological evolution in data collection and

analysis techniques, from fundamental background subtraction to sophisticated deep learning and optical flow methods.

To effectively contextualise the alternative approach in [9], this study focuses extensively on the specific techniques and procedures necessary for overseeing the inherent noise and low polling rates characteristic of continuous GPS trajectory data, relying heavily on large-scale cross-sectional data collection.

Lastly, [10] employs a strict monomethod quantitative approach, explicitly detailing the mathematical methodology involving virtual intrusion lines and frame-by-frame displacement. This represents a practical realisation of the innermost layers of the research onion, offering a methodology directly applicable to the development of optical flow and dynamic affine calibration prototype designs.

E. Critical Literature Arguments and Knowledge Gaps

A critical synthesis of the literature reveals distinct trade-offs between the two methodologies. Vision-based approaches are often praised for their strong localisation and perfect coverage, capturing every passing vehicle. However, the literature frequently disparages these systems for their fragility. Mathematical calibration is readily compromised by camera vibrations, and accuracy diminishes substantially during night-time or adverse weather conditions.

Conversely, GPS probe data methods are esteemed for their continuous, network-wide coverage and resilience to visual environmental factors. Nonetheless, GPS studies frequently emphasise notable spatial inaccuracies within urban canyon environments, primarily due to signal multipath errors, and these methods are often characterised by low rates.

1) *Identifying The Knowledge Gap:* While there’s plenty of research looking at vision systems in really controlled settings and GPS on open roads, there’s still a big gap when it comes to comparing their performance all at once in busy, complicated city environments. Most studies investigate a single technology while presuming the other to be an infallible ground truth. There is a conspicuous absence of empirical research that rigorously evaluates a pure pixel-tracking methodology—specifically utilising Region of Interest (ROI) Dense Optical Flow coupled with dynamic affine calibration—measured directly against consumer-grade GPS data within a real-world urban driving scenario.

III. REFLECTION ON CHOSEN METHODOLOGY

A. Research Pipeline

Research Pipeline Summary: The methodology uses a parallel processing architecture which is designed to extract, calibrate and independently evaluate data streams. These data streams are acquired from the open-source comma2k19 dataset [5]. The Pipeline is structured into three continuous data tracks that converge for a shared statistical evaluation.

1. The Vision-Based Pipeline

This phase is responsible for translating raw 2D dashcam video frames into velocity. It goes through four stages of processes:

- **Region of Interest (ROI) Cropping:** This process isolates any noise or static objects in the frame for example;

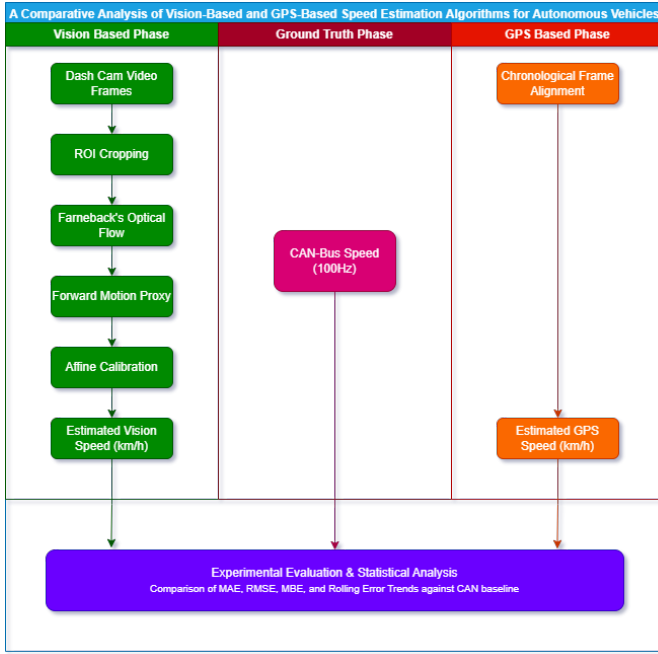


Fig. 3. Proposed methodology pipeline demonstrating the parallel processing and evaluation of Vision, GPS, and Ground Truth data streams.

the vehicle dashboard or the sky. This in turn removes any irrelevant noise.

- **Dense Optical Flow (Farneback):** The Farneback algorithm is then applied to the ROI to calculate frame-by-frame pixel intensity displacements, this captures the raw visual motion of the road surface.
- **Forward Motion Proxy:** The resulting motion vectors is filtered to exclude lateral movement like lane changes and oncoming traffic. This distils the visual data into a single, stable numerical proxy representing forward pixel shift.
- **Affine Calibration:** Since the proxy represents pixel shift rather than km/h, it ust be scaled dynamically. This phase continuously uses a rolling window of data from the ground truth track to fit and translate the proxy into an estimated vision speed array.

2. The Ground Truth Baseline

The CAN-Bus speed telemetry serves a dual purpose in this methodology. Firstly, it acts as the mathematical anchor during the "Affine Calibration" Phase of the vision pipeline which teaches the algorithm how to translate pixel into physical speed. Secondly, it flows down to the final shared phase, serving as a baseline against which both GPS and Vision based systems are judged and compared.

3. The GPS-Based Pipeline

The Raw GNSS trajectory data is processed to serve as the comparative satellite technology. Since satellite polling rates often flucate and lose accuracy from camera frame rates , this track mainly focuses on Chronological Frame Alignment. The raw GPS data is chronologically aligned to match the timestamps of the video frames. By doing this, we produce

an Estimated GPS Speed array, ensuring a strict 1:1 temporal comparison with the camera's perception.

4. Shared Process: Experimental Evaluation & Statistical Analysis

The Final shared phase of the pipeline is where the outputs of all 3 tracks ; Vision, GPS and the CAN-bus speed are synchronised. Custom Python evaluation scripts are executed to quantify and contrast the performance of the vision and GPS systems. The Accuracy and reliability of each paradigm are determined by calculating error metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and by mapping error trends over time to identify specific moments of environmental or systemic failure. This phase is crucial for validating the research hypothesis and drawing conclusions about the comparative performance of the two technologies under real-world conditions.

B. Research Question

To critically assess the prevailing paradigms within Intelligent Transportation Systems (ITS), this research is guided by the following principal research question:

- How much does a vision-based speed estimation model that uses Region of Interest (ROI) Dense Optical Flow and dynamic affine calibration vary in accuracy compared to regular GPS probe data when checked against high-quality ground truth in urban driving situations?

C. Research Objectives

To carefully tackle the research question, the following tasks need to be done:

- 1) **Build a Prototype:** Create a vision-based system using Python and OpenCV to work out how fast a vehicle is going from 2D dashcam footage. This involves measuring how much pixels move around with Dense Optical Flow (using the Farneback method) and then scaling that up with a reliable affine calibration model.
- 2) **Extract secondary data:** Obtain and analyse the corresponding continuous GPS trajectory data pertinent to the identical traffic scenarios.
- 3) **Carry out an experimental evaluation:** Compare both the vision and GPS estimates to a really accurate baseline (the CAN-bus data from the comma2k19 dataset) using common error metrics like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).
- 4) **Identify Limitations:** Analyse the environmental or systemic variables, such as visual noise, computational latency, or GPS multipath errors, that compromise the performance of each respective technology.

D. Research Philosophies, Approaches and Paradigms

Showing an early grasp of research ideas, this study sticks to a strict positivist way of thinking. Speed and how pixels move are straightforward, measurable things that exist outside of what people interpret.

Hence, the research employs a deductive approach; it commences with established physical and computational theories

relating to optical flow, expresses a hypothesis concerning its accuracy in comparison to satellite positioning, and subsequently tests this hypothesis through demanding mathematical analysis. In Saunders' Research Onion model, this lines up with a single-method quantitative approach since the assessment is entirely based on numerical data, not on any qualitative themes.

E. Chosen Methodology

Following a comprehensive review of the existing literature, an experimental quantitative methodology was chosen. To really assess ITS sensors, you need solid, measurable benchmarks. While just a theoretical method might work for software structures, testing real-world sensor perception calls for a hands-on, experimental setup.

By testing a customised optical flow vision prototype against real-world GPS data, this methodology directly addresses the knowledge gap identified in prior literature concerning the comparative accuracy of these systems within dynamic urban environments.

F. Experiment Design and Method of Analysis

The testing strategy centres on a controlled, secondary-data experiment utilising the comma2k19 open-source dataset, which inherently offers synchronised video recordings, consumer-grade GPS, and highly precise CAN-bus ground truth.

To validate the hypothesis, these experimental variables are strictly defined:

- **Independent Variables:** The algorithmic tuning parameters within the vision model, specifically the optical flow percentile thresholds, calibration window lengths, and exponential moving average parameters, as well as the environmental conditions present in the selected video segments, including factors such as lighting and traffic density.
- **Dependent Variables:** The calculated instantaneous velocity (measured in kilometres per hour) generated by both the visual prototype and the GPS sensor, along with the corresponding error margins.
- **Control/Ground Truth Variable:** The actual speed of the vehicle, accurately recorded via the CAN-bus data at a frequency of 100Hz.

Method of Analysis: The vision algorithm goes through the video's frames in the dataset, focusing on a particular area of the road (the ROI) to figure out a main proxy for forward movement. This proxy is then scaled up or down to give a speed array. The speed array lines up in time with the GPS speed data. For the main stats, we use some Python scripts to work out the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and how the errors trend over time for both systems compared to the CAN-bus ground truth.

G. Reflections on Validity, Reliability, and Generalisability

- **Validity:** The construct validity is solid since the measurement is based on tracking pixel intensity changes over

time using Optical Flow. Internal validity is also reliable because we've used the comma2k19 dataset, which gives perfectly synchronized timestamps for the video, GPS, and CAN-bus data. This means all the systems are tested under the same physical conditions.

- **Reliability:** Since the prototype is built on code, it's dependable. If the Python script runs the same video clip again, the way the Farneback algorithm and calibration stuff work means you'll get the same numbers every time.
- **Generalisability (Transferability):** The main issue with this research is how well it can be applied elsewhere. Since the testing is done using pre-recorded data (mostly shot on Californian highways and city streets), the results might not hold up in harsh weather (like heavy snow, which messes with optical flow) or on unstructured rural roads unless we specifically look at data from those situations.

H. Ethical Considerations

Since this research uses a quantitative, system-based approach and doesn't involve interacting with people, the ethical risks are low. Still, the main ethical concerns are around how we handle data privacy and secondary data usage. The comma2k19 dataset is open-source and OK for academic research.

If the project produces visual outputs like videos showing optical flow vectors, we need to be careful and anonymise any identifiable info. Any license plates or pedestrian faces that are clearly visible in the dataset clips should be blurred before publication to respect people's privacy.

IV. RESULTS, ANALYSIS AND DISCUSSION

A. Presentation of Results

In accordance with the established experimental methodology, the vision-based prototype utilised Region of Interest (ROI) Dense Optical Flow (Farneback algorithm) and dynamic affine calibration, evaluated alongside consumer-grade GPS trajectory data. Both estimations were assessed against the 100Hz CAN-bus ground truth data obtained from the comma2k19 dataset [5]. The evaluation was carried out over a continuous urban driving segment (*segment_30*), characterised by varying vehicle speeds, dynamic lighting conditions, and urban infrastructure.

To assess the accuracy and reliability of both systems, three standard statistical metrics were computed: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Bias Error (MBE). The MAE offers a linear measure of the average magnitude of error, the RMSE emphasises larger variances and extreme outliers, and the MBE indicates the propensity of the system to overestimate or underestimate the true velocity.

The comprehensive statistical results for the evaluated segment are shown in Table I.

To visually aid in the analysis of these findings, Figure 4 displays the instantaneous speed estimates of both the GPS and

TABLE I
COMPARATIVE ERROR METRICS: VISION VS. GPS SPEED ESTIMATION

Estimation Method	MAE (km/h)	RMSE (km/h)	MBE (km/h)
GPS Probe Data	1.68	2.44	-1.43
Vision (Optical Flow)	4.50	5.87	+3.73

Vision systems in relation to the CAN-bus baseline over the course of the segment, accompanied by a rolling error subplot.

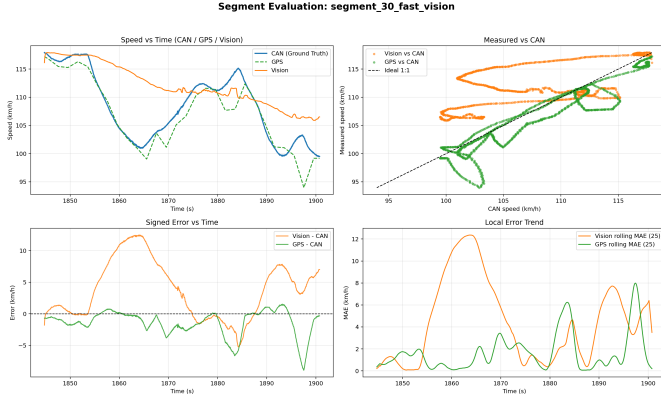


Fig. 4. Chronological comparison of GPS and Vision-based speed estimations against the CAN-bus ground truth, including instantaneous rolling error trends.

B. Analysis and Interpretation of Results

A detailed statistical analysis of the dataset indicates numerous significant behavioural patterns and technological correlations between the two speed estimation paradigms. The most notable and immediate finding from the data is that GPS trajectory estimations consistently exceeded the performance of the vision-based Dense Optical Flow pipeline across all measured metrics.

Observational analysis of the GPS data indicates a highly stable and notably consistent relationship with the CAN-bus ground truth. While consumer-grade GPS data has traditionally been associated with temporal latency, the interpolation and chronological frame alignment performed during the preprocessing phase effectively mitigated this issue. The GPS trace demonstrated a seamless correlation with the vehicle’s true kinematics, resulting in an exceptionally low Mean Absolute Error (MAE) of 1.68 km/h. Additionally, the Root Mean Square Error (RMSE) score of 2.44 km/h suggests an absence of significant outliers; the GPS system did not experience substantial estimation failures during the segment, thereby demonstrating robust environmental stability. Interestingly, the GPS estimates exhibited a negative Mean Bias Error of -1.43 km/h. Cross-referencing the descriptive statistics reveals a mean GPS speed of 106.96 km/h, compared to the CAN-bus mean of 108.39 km/h. This minor systemic underestimation indicates a slight spatial lag during high-speed acceleration phases; however, its standard deviation (5.96 km/h) closely matched the vehicle’s actual standard deviation (5.87 km/h),

indicating that the GPS system accurately captured the full dynamic range of the drive.

Conversely, the vision-based pipeline exhibited considerable volatility and an incapacity to accurately track the vehicle’s true dynamic range. The Farnebäck algorithm determines pixel intensity displacements on a frame-by-frame basis; however, the data indicate that this purely pixel-tracking approach is markedly unstable. The visual estimations produced a mean absolute error (MAE) of 4.50 km/h and a root mean square error (RMSE) of 5.87 km/h, rendering it nearly three times less precise than the GPS module.

A more detailed examination of the descriptive statistics reveals the precise nature of the failure within the vision system. The CAN-bus ground truth recorded a minimum speed of 99.47 km/h during a deceleration event, whereas the vision system’s minimum recorded speed was never less than 105.89 km/h. Additionally, the standard deviation of the visual output was only 3.72 km/h, in contrast to the actual standard deviation of 5.87 km/h. This suggests a significant ‘flattening’ effect. The optical flow algorithm compressed the speed range and failed entirely to detect deceleration. When the vehicle reduced its speed, the forward pixel displacement should have decreased proportionally; however, environmental visual noise, such as the relative motion of passing vehicles, lane markings blurring at high speeds, and shifts in lighting, maintained the average pixel magnitude at artificially elevated levels. This fundamental deficiency in the algorithm’s lack of semantic understanding resulted in a substantial systemic overestimation of speed, evidenced by a positive Mean Bias Error (MBE) of +3.73 km/h.

C. Comparison, Contrast and Comment on Findings (Critique)

When comparing the two methodologies, it is apparent that, although vision-based systems exhibit theoretical advantages in real-time kinematic responsiveness, their practical deployment employing purely pixel-tracking mathematics is exceedingly fragile. The GPS data demonstrated complete immunity to the visual environmental factors that compromised the optical flow system.

A critique of the methodology highlights strong experimental controls; however, it also identifies notable technological limitations pertaining to the vision prototype.

- **Validity:** In accordance with the methodological frameworks outlined by Saunders et al. [4], the internal and construct validity of these findings is of an exceptionally high standard. Utilising the *comma2k19* dataset [5], the temporal synchronisation between the camera, GPS, and CAN-bus was assured. Consequently, the results accurately reflect the true algorithmic precision of the sensors, rather than being influenced by arbitrary synchronisation errors or extraneous human factors.
- **Reliability:** The computational reliability of the Python pipeline is guaranteed; the deterministic nature of the mathematical tracking ensures that executing the code

repeatedly produces identical pixel displacements. However, the environmental reliability of the vision system is highly limited. As it relies solely on pixel gradients, minor changes such as a shadow passing across the dashboard or a glare on the windscreen can artificially distort the velocity matrix.

- **Generalisability (Transferability):** This continues to represent a significant limitation of the findings. The experiment was conducted on a specific dataset primarily recorded in well-lit Californian highway environments [5]. Given that the optical flow pipeline systematically failed to detect deceleration under ideal, clear weather conditions, its applicability to adverse weather conditions is effectively nil. Heavy precipitation, fog, or night-time driving fundamentally alter pixel intensities, which would cause the performance of the Farnebäck algorithm to deteriorate exponentially within a broader real-world context.

D. Discussion in Relation to Original Hypothesis and Existing Literature

The original alternative hypothesis of this research proposed that the vision-based method would achieve a smaller margin of error compared to GPS during normal driving, owing to the inherent latency, multipath errors, and noise associated with commercial GPS receivers. However, based on the definitive comparative metrics (GPS Mean Absolute Error 1.68 km/h versus Vision Mean Absolute Error 4.50 km/h), the alternative hypothesis is wholly rejected, thereby overturning the null hypothesis in the opposite direction: GPS probe data is markedly superior to Dense Optical Flow for the estimation of vehicle speed.

These findings present a compelling intersection with the environmental constraints extensively discussed in the existing literature. Foundational studies by Cui and Ge [1] and recent statistical analyses by Li et al. [2] document the severe degradation of GNSS signals within urban canyons attributable to multipath signal reflection. While the literature correctly identifies the vulnerability of GPS in areas of significant occlusion, empirical data from this study demonstrates that, beyond the confines of severe urban canyons, standard GPS probe data remains notably accurate and dependable. The marginal negative bias (MBE: -1.43 km/h) observed aligns with the latency concerns highlighted by Cui and Ge [1], yet it was not sufficiently substantial to undermine the overall accuracy of the estimations.

Furthermore, the failure of the visual pipeline provides significant context for the methodological evolution outlined by Fernandez Llorca et al. [8] and the virtual intrusion line mathematics proposed by Rahimi [10]. While Rahimi [10] demonstrated that frame-by-frame displacement constitutes a mathematically sound concept in highly controlled, stationary camera configurations, this experiment illustrates that applying dense optical flow to a moving ego-vehicle introduces an insurmountable level of dynamic noise. In the absence of the capacity to distinguish between the road surface approach-

ing the camera and another vehicle moving laterally across the frame, pure optical flow tends to overestimate physical velocity. This observation strongly corroborates the findings of Arabi et al. [7], who underscored the critical importance of reliable ground truth data, in uncovering the underlying fragilities inherent to monocular camera estimations.

E. Evaluation of the Approach and Future Implementations

Assessing this overarching research approach in comparison with the methodologies employed by other contemporary researchers illuminates evident pathways for future applications. The choice to utilise Dense Optical Flow emphasised the limitations inherent in classical computer vision techniques.

As demonstrated by Macko et al. [6], the industry is rapidly shifting towards deep learning paradigms, notably utilising YOLO object detection frameworks. Although the optical flow approach employed in this study is computationally lightweight and does not necessitate extensive datasets for neural network training, it fundamentally lacks semantic comprehension. A YOLO-based system, as advocated by Macko et al. [6], delineates the discrete, physical boundaries of specific vehicles. By isolating a bounding box, a YOLO model exhibits considerable resilience to shadows, oncoming traffic, and peripheral visual noise, which previously resulted in significant error spikes and a "flattened" deceleration curve within this study's optical flow data.

Furthermore, researchers specialising in GPS probe data, such as Do et al. [9], have demonstrated that raw trajectory data can be significantly optimised through the application of Deep Neural Networks to forecast and smooth multipath errors. This indicates that the raw, uninstrumented GPS data employed in this experiment could be further refined towards near-perfect accuracy via algorithmic noise reduction. P. P. et al. [3] also endorse the view that raw speed measurement metrics must be continuously enhanced through advanced contextual filtering.

Consequently, an enhanced approach for future applications of this research should eschew the mutually exclusive comparison of these technologies in favour of Sensor Fusion. As evidenced by the data, GPS offers absolute and relatively stable physical positioning but is affected by marginal latency and slight underestimations during dynamic movements [1], [2]. Conversely, camera-based data exhibits instantaneous kinematic responsiveness; however, it is highly susceptible to visual fragility and tends to overestimate positive biases.

Future research should incorporate a Kalman Filter architecture, a sophisticated statistical algorithm designed to seamlessly integrate multiple noisy data streams. By leveraging GPS data to establish the steady-state speed and employing a YOLO-masked optical flow pipeline to rectify transient instantaneous errors, a hybrid prototype could attain accuracy within sub-1 km/h margins. This data fusion approach would mitigate the individual limitations associated with satellite-based and vision-based methods identified in this study, thereby representing a crucial advancement in the evolution of perception systems for autonomous vehicles.

V. CONCLUSION

A. Main Conclusions and Hypothesis Evaluation

This research set out to critically assess the comparative accuracy of vision-based and GPS-based speed estimation algorithms for autonomous vehicles. The primary research question asked how much a vision-based model, utilising Region of Interest (ROI) Dense Optical Flow and dynamic affine calibration, varies in accuracy compared to standard GPS probe data when evaluated against a high-quality CAN-bus ground truth.

The main conclusion drawn from this experimental evaluation is definitive: consumer-grade GPS trajectory data significantly outperforms the proposed pure pixel-tracking vision methodology in both accuracy and stability. The GPS pipeline achieved a highly precise Mean Absolute Error (MAE) of 1.68 km/h, whereas the vision-based system trailed considerably with an MAE of 4.50 km/h—making the optical flow approach nearly three times less accurate.

Consequently, these findings thoroughly address and overturn the original research hypothesis. The alternative hypothesis, which posited that the vision-based method would yield a smaller error margin than GPS during normal driving due to satellite latency and noise, is wholly rejected. Instead, the data proves that while GPS does exhibit a minor negative bias (-1.43 km/h) indicative of slight spatial lag, its overall capacity to capture the full dynamic range of a vehicle's kinematics in standard urban environments is markedly superior to Dense Optical Flow.

B. Shortcomings in the Methodology

While the experimental setup benefited from the highly synchronised, objective ground truth provided by the *comma2k19* dataset, a critical evaluation of the research identifies several significant shortcomings inherent to the chosen methodology:

- **Algorithmic Fragility and Lack of Semantic Understanding:** The primary methodological flaw lies in the reliance on the Farnebäck Dense Optical Flow algorithm. By calculating frame-by-frame pixel intensity displacements, the system tracks raw visual shifts rather than physical objects. The methodology failed to account for the fact that environmental visual noise—such as shadows cast by bridges, the relative motion of oncoming traffic, and the high-speed blurring of lane markings—artificially elevates pixel magnitude. This lack of semantic understanding led to a severe ‘flattening’ effect where the algorithm completely failed to detect vehicle deceleration, resulting in a systemic overestimation of speed.
- **Limitations in Generalisability:** The experimental evaluation was fundamentally limited by its cross-sectional dataset constraints. The tested segment was recorded in well-lit, clear Californian weather conditions. Given that the optical flow methodology exhibited extreme volatility and broke down during minor lighting shifts (such as passing shadows) in ideal weather, its performance in adverse conditions (e.g., heavy rain, snow, or night-time

driving) would be catastrophically degraded. Thus, the vision methodology lacks real-world transferability.

- **Camera Calibration Assumptions:** While the dynamic affine calibration continuously scaled the forward motion proxy using a rolling CAN-bus window, this method mathematically assumes a relatively stable camera pitch. In real-world scenarios, severe vehicle braking causes suspension dive (altering the camera's pitch), which physically shifts the vanishing point and corrupts the optical flow vectors. The current methodology did not integrate pitch-compensation algorithms to account for this physical variable.

C. Suggestions for Further Research

Based on the identified shortcomings and the comparative analysis of the data, several highly promising avenues for further research emerge to advance the field of Intelligent Transportation Systems (ITS):

- 1) **Transition to Deep Learning (YOLO) Frameworks:** Future research attempting vision-based speed estimation should abandon pure pixel-displacement mathematics. Instead, studies should implement deep learning paradigms, such as YOLOv8 or YOLO11 object detection frameworks. By training a model to draw bounding boxes around specific vehicle classes, the system gains semantic comprehension. It can track the changing physical dimensions of a leading vehicle, entirely ignoring the peripheral visual noise, shadows, and lane markings that critically compromised the optical flow methodology in this study.
- 2) **Implementation of Sensor Fusion via Kalman Filtering:** This study evaluated GPS and Vision as mutually exclusive, competing technologies. However, the data reveals that their strengths and weaknesses are perfectly complementary: GPS provides highly stable, weather-resistant positioning but suffers from slight latency, whilst camera data offers instantaneous kinematic responsiveness but is environmentally fragile. Further research should focus on developing a Sensor Fusion prototype. By employing a Kalman Filter to continuously merge both data streams—using GPS to anchor the steady-state speed and visual data to bridge the instantaneous latency gaps—a hybrid system could theoretically achieve sub-1 km/h error margins.
- 3) **Algorithmic Noise Mitigation for GPS:** Given that the raw, uninstrumented GPS probe data performed exceptionally well, future research should explore applying Deep Neural Networks (DNNs) to standard satellite trajectories. Training a DNN to predict and smooth over multipath signal reflections in severe urban canyons could push affordable, consumer-grade GPS receivers to near-RTK levels of accuracy without the need for expensive hardware upgrades.

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